Deep Learning Pipeline for Liver Vessel Segmentation

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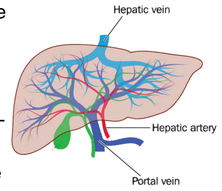
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Introduction

Importance of Liver Vasculature in Transplant Surgery: Precise mapping of liver vessels is critical for successful liver transplants. The liver's complex vascular structure includes hepatic arteries, portal veins, and hepatic veins, each essential for the organ's function. Surgeons need exact vessel maps to make safe incisions and ensure the transplanted liver functions properly.

Challenges with Manual Segmentation: Manual segmentation of liver vessels is time-consuming and prone to errors, depending heavily on the radiologist's experience. This can lead to inconsistent surgical planning, highlighting the need for more efficient and reliable methods.



Liver Anatomy

Difficulties of Integrating AI in Clinical Workflows:
Integrating AI solutions for liver vessel segmentation into existing clinical workflows is challenging requiring alignment with healthcare standards and practices to ensure AI tools enhance rather than disrupt clinical operations.

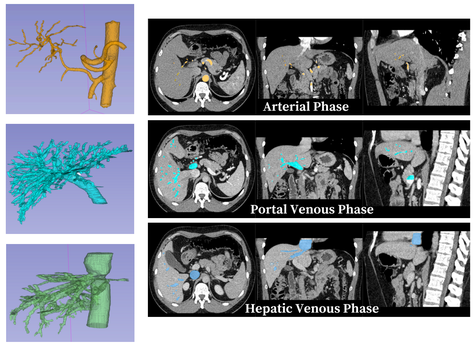
Objective

The aim of this project was to develop and deploy a deep learning pipeline that can segment hepatic vessels from abdominal CT scans. The project also primarily uses open source software for all of its development and deployment, to contribute to advances in radiology and medical AI.

Phase 1: Development and Initial Testing:
The initial phase, involved pre-processing the training data and training a convolutional neural network (CNN) to segment the liver vessels from triple-phase CT scans, aiming achieving a relatively high accuracy as measured by the DICE coefficient.

Phase 2: Deployment and Workflow Integration:
The subsequent phase aimed to fully integrate the pipeline into the clinical workflows, ensuring compliance with DICOM standards and interaction with PAC systems.

Methods and Results

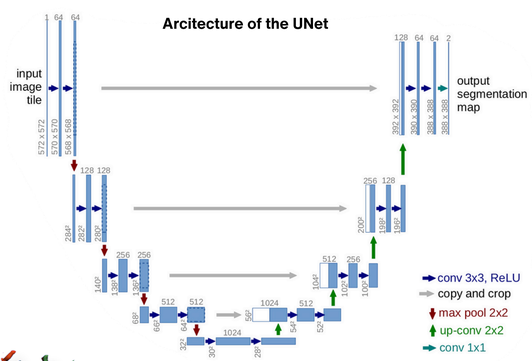


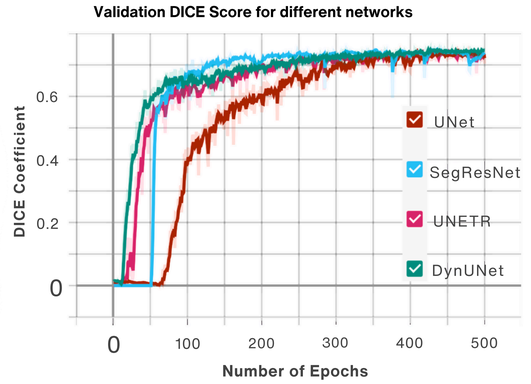
Data Acquisition and Preparation:

- We acquired 26 triple-phase liver CT scans from Cleveland Clinic Abu Dhabi. Each scan captures different phases of blood flow through the liver, crucial for detailed imaging of hepatic vessels. The data underwent preprocessing - included converting DICOM images to NIfTI which is needed for training a neural network.

Training and Model Development:

- CNN architectures such as UNet, DynUNet, SegResnet and DynUNet were chosen for their efficacy in medical image segmentation. Training involved 500 epochs. Given the small size of the dataset, we applied techniques like on-the-fly data augmentation—including rotations, cropping, and intensity shifts—to enhance the model's ability to generalize from the training data to unseen data.



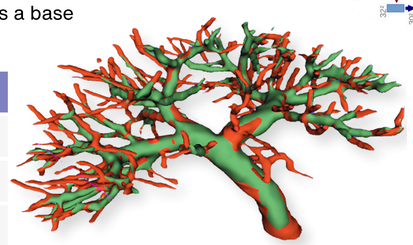


Performance Evaluation:

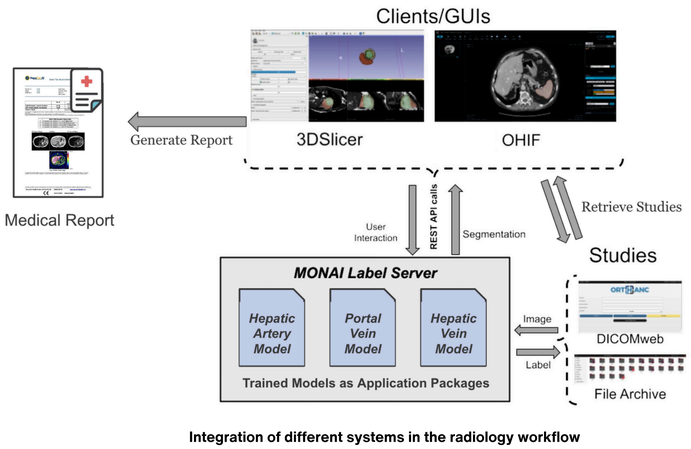
- The models proficiencies were notable in accurately capturing the primary hepatic vessels and their initial bifurc tions, as shown in the figure. However, it encountered difficulties in segmenting smaller, more intricate vessels.
- Despite all networks performing similarly, the model trained using UNet was selected for ongoing development due to its proven effectiveness in medical image segmentation. Its performance, demonstrated in our trials, and in other medical imaging tasks, makes it a reliable choice as a base model to develop upon.

Phase	DICE Score
Hepatic Arterial	0.67
Portal Venous	0.75
Hepatic Venous	0.70

Performance with UNet for the three phases


Predicted (Green) and Ground Truth (Red) for trained model

Deployment

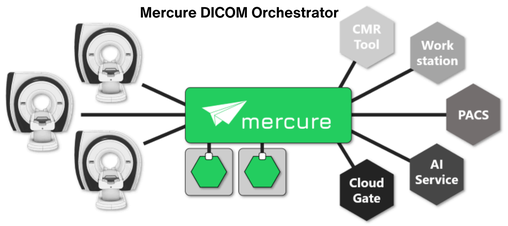


Integration of different systems in the radiology workflow

Workflow Integration: The deployment architecture was designed to seamlessly integrate with the clinical workflows of various healthcare facilities. This involved connections between critical components such as the radiologist's viewer, hospital PACS systems, and the processing server running the AI model, which utilizes MONAI Label for advanced medical imaging processing.

DICOM Protocol Utilization: The entire system leverages the DICOM protocol to ensure compatibility and standardization in the handling, storage, and transmission of medical images. This standard protocol facilitates the integration of our AI tools with existing medical imaging infrastructures, maintaining consistency across different systems.

Mercure as Orchestrator: Mercure was employed as a robust orchestration tool to manage the flow and queue of imaging data across the network. It applies different processing modules based on predefined rules and routes the processed data back to the respective PACS systems. This orchestration supportsreal-time updates and efficient handling of data requests, enhancing the overall efficiency of the deployment.



Challenges and Future Work

Challenges:
Limited Data Variability: The restricted variety in training datasets hindered the model's learning capability across diverse anatomical variations, impacting its generalizability and accuracy.
Complex Vascular Structures: Achieving clinically viable segmentation accuracy was challenging due to the intricate nature of liver vascular structures.
Software System Integration: Ensuring integration of various software systems with existing clinical infrastructures and databases was complex and posed significant challenges.

Future Work:
Dataset Expansion: Plans include gathering more diverse datasets from various healthcare facilities or generating synthetic datasets with techniques like GANs.
Enhancing Model Accuracy: Exploring ensemble methods and other advanced machine learning strategies to better handle the complexities of liver vascular structures.
Interactive Interface Development: Creating a user-friendly interface for active learning that allows clinicians to provide real-time feedback, enhancing model adaptability and customization to clinical needs.

References

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